PROTOTYPES FOR HUMANITY 2026 / COMPETITION PACKAGE

BRINEWATCH

See the plume. Target the evidence.

An underwater monitoring prototype that combines forward-looking imaging sonar, CT sensing, adaptive sampling and an uncertainty-aware digital twin for desalination outfalls.

INITIAL PROTOTYPE OR MODEL

Simulation-backed prototype / real custom-HoloOcean mission evidence

01 / PROBLEM

Sparse measurements can miss the spatial boundary

BrineWatch adds a repeatable screening layer before certified/manual sampling is deployed.

Current monitoring limitation

Fixed stations and limited CT casts can confirm conditions only where they were placed. A moving, bottom-hugging plume can remain spatially under-resolved, while a conventional spatial vessel survey can be costly to repeat.

BrineWatch value

More informative spatial evidence under constrained survey time - and a map that helps direct certified/manual sampling toward the locations that matter. The prototype does **not** claim to universally replace accredited monitoring.

SAME-DAY MAP

TARGETED FOLLOW-UP

SITE HISTORY

One mission links infrastructure to environmental evidence

The planned payload is modular: owned BlueROV2 + owned Omniscan 450 FS + next-step CT integration.

1
LOCATE
Omniscan 450 FS
Confirm the diffuser from several aspects.

2
SENSE
CT payload
Sample salinity and temperature along the route.

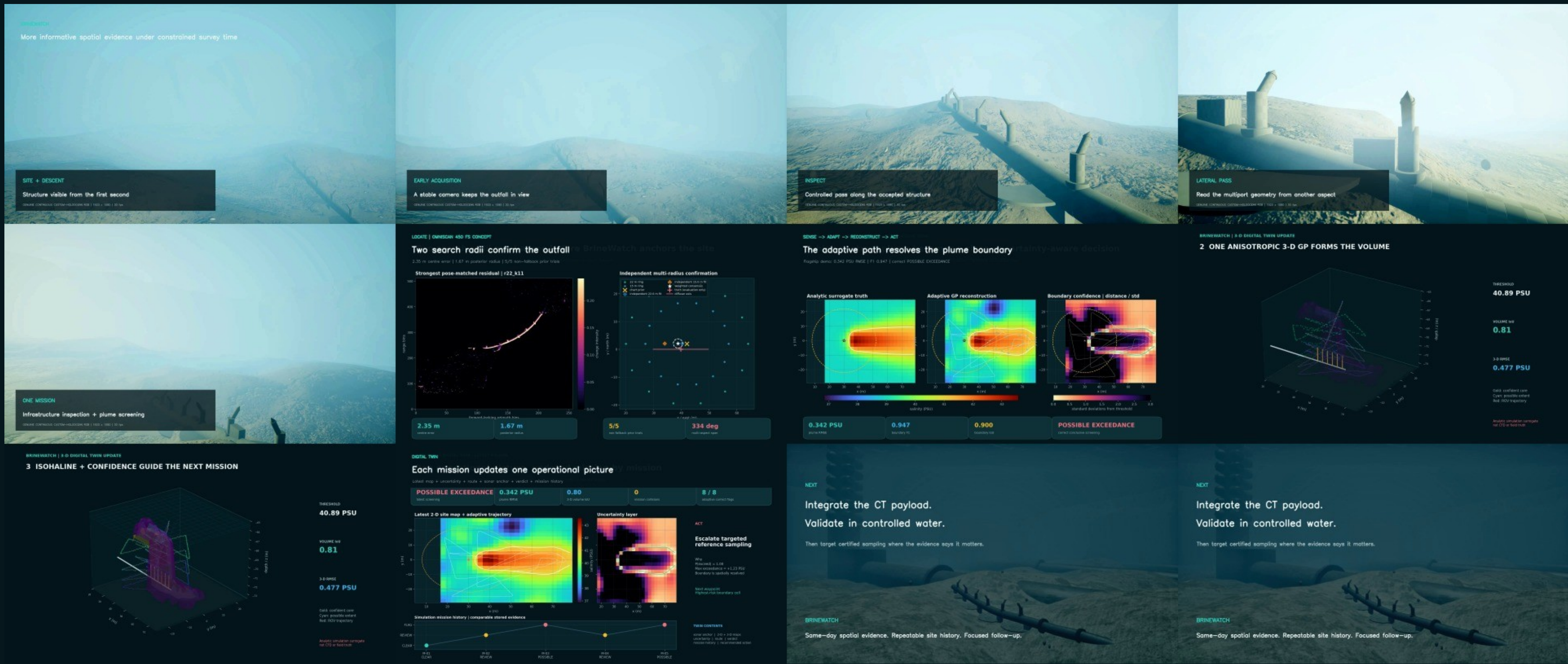
3
ADAPT
Information gain
Spend travel budget near signal and uncertainty.

4
RECONSTRUCT
2-D + 3-D GP
Store the estimate and confidence, not only points.

5
ACT
Three-state screen
CLEAR, REVIEW or POSSIBLE EXCEEDANCE.

The ROV completed a collision-free in-engine mission

The accepted outfall geometry was not redesigned. Vehicle motion and sonar are native simulation; the plume remains analytic.



CUSTOM ENGINE	TRAVEL	SAFETY	CLEARANCE	SCREENING
564	395 m	0	1.85 m	REVIEW
CT samples	in-engine route	collisions	measured minimum	honest abstention

The 2.0 m standoff target was undershot by 0.15 m during dynamics but no collision occurred. This custom-engine run is locomotion/sensing evidence, not the headline reconstruction case.

Multi-radius confirmation makes the sonar anchor defensible

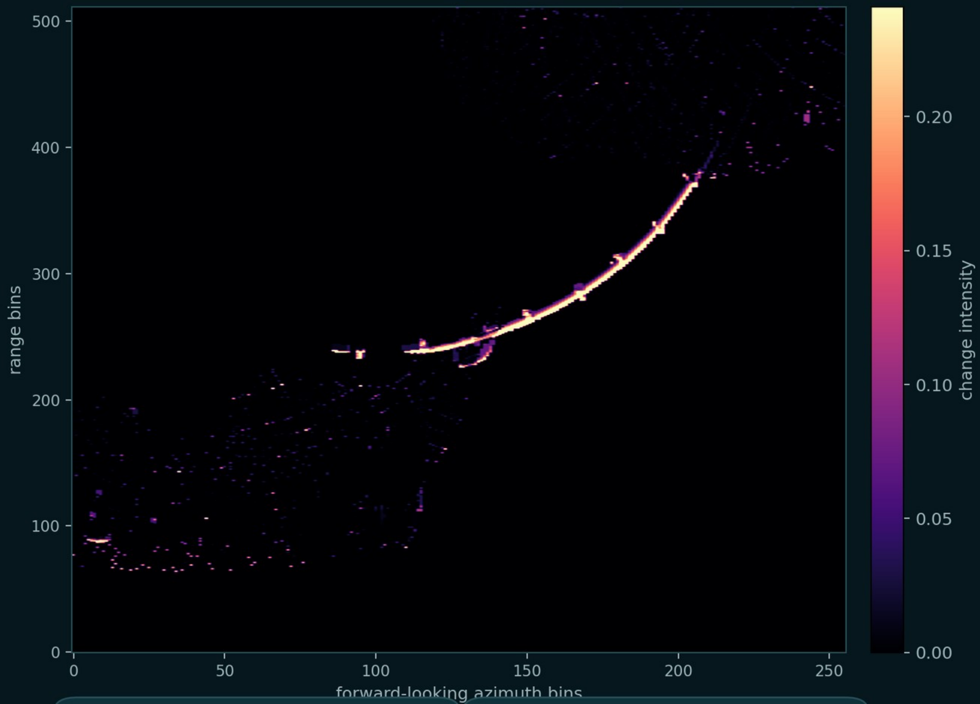
The planned and simulated sensing concept is consistently presented as forward-looking imaging sonar.

LOCATE | NATIVE HOLOOCEAN SONAR

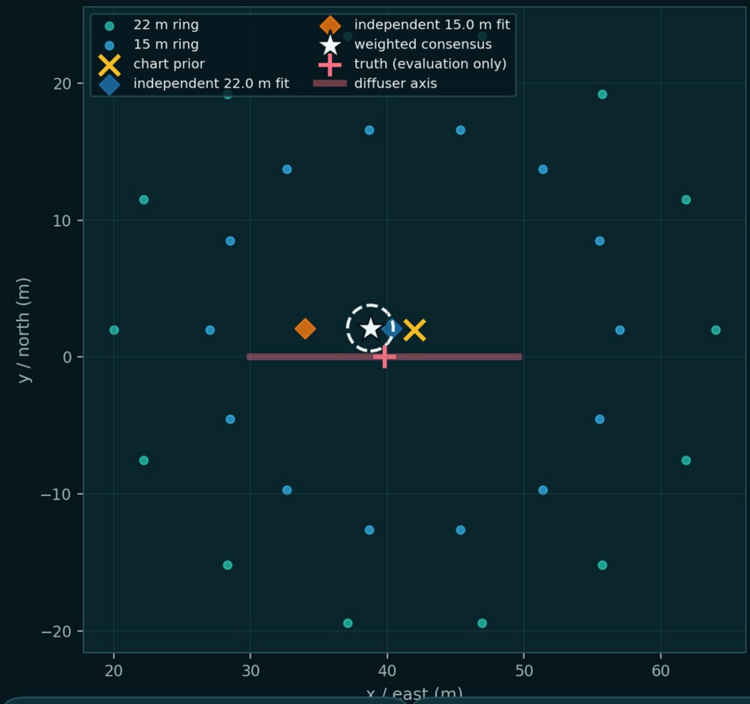
Two search radii must agree before BrineWatch anchors the site

Pose-matched background subtraction rejects persistent terrain clutter; no oracle input or silent fallback.

Strongest pose-matched residual | r22_k11



Independent multi-radius confirmation



2.35 m
centre error

1.67 m
posterior radius

5/5
non-fallback prior trials

334 deg
multi-aspect span

Nominal two-radius inverse-uncertainty consensus: 2.35 m centre error, 1.67 m posterior radius and 334 deg aspect span. Five chart-prior perturbation fits remained non-fallback; their post-hoc median error was 4.24 m. No oracle coordinate entered localization.

05 / 2-D RESULT

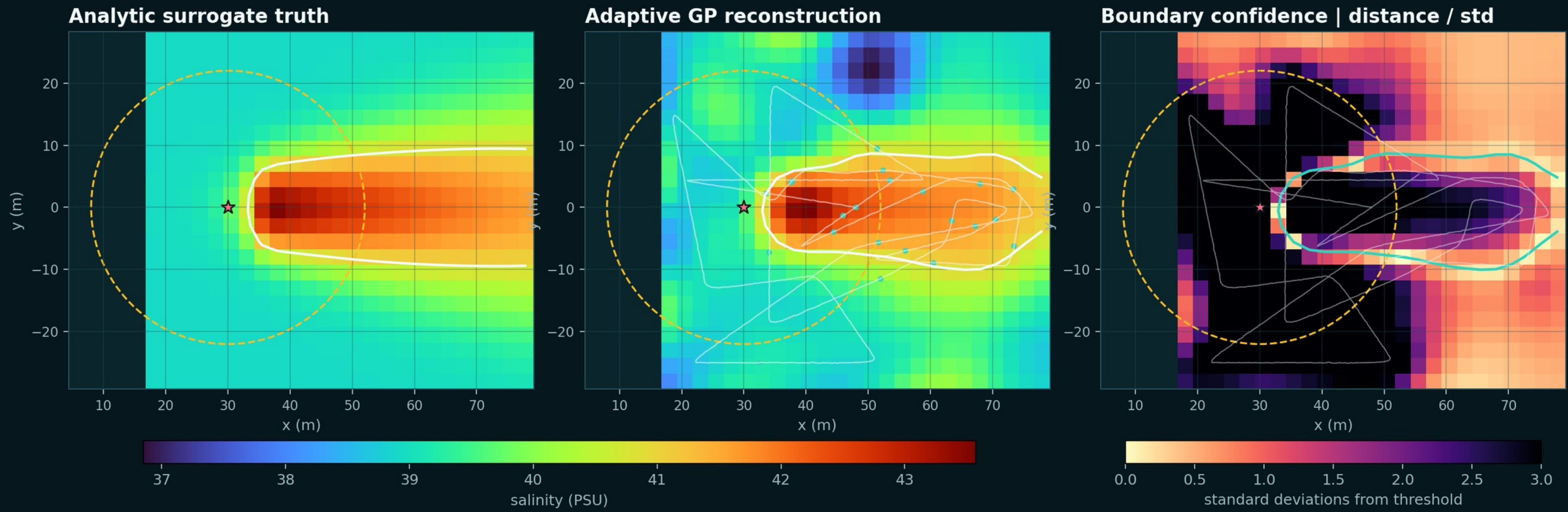
A clear plume supports a correct conclusive screen

Demo-optimised high contrast is allowed for communication - and explicitly labelled as an analytic simulation surrogate.

SENSE -> ADAPT -> RECONSTRUCT -> ACT | FLAGSHIP DEMO

A high-contrast plume becomes a clear, uncertainty-aware decision

Demo-optimised analytic surrogate, explicitly not CFD or field truth. Same BrineWatch workflow and safety logic.



0.342 PSU	0.947	0.900	POSSIBLE EXCEEDANCE
plume RMSE	boundary F1	boundary IoU	correct conclusive screening

Flagship scenario: static tide, compact 60 m x 56 m site, 520 m budget and high salinity contrast. Result: 591 samples, 0.342 PSU plume RMSE, boundary F1 0.947, boundary IoU 0.899 and correct POSSIBLE EXCEEDANCE vs non-compliant surrogate truth.

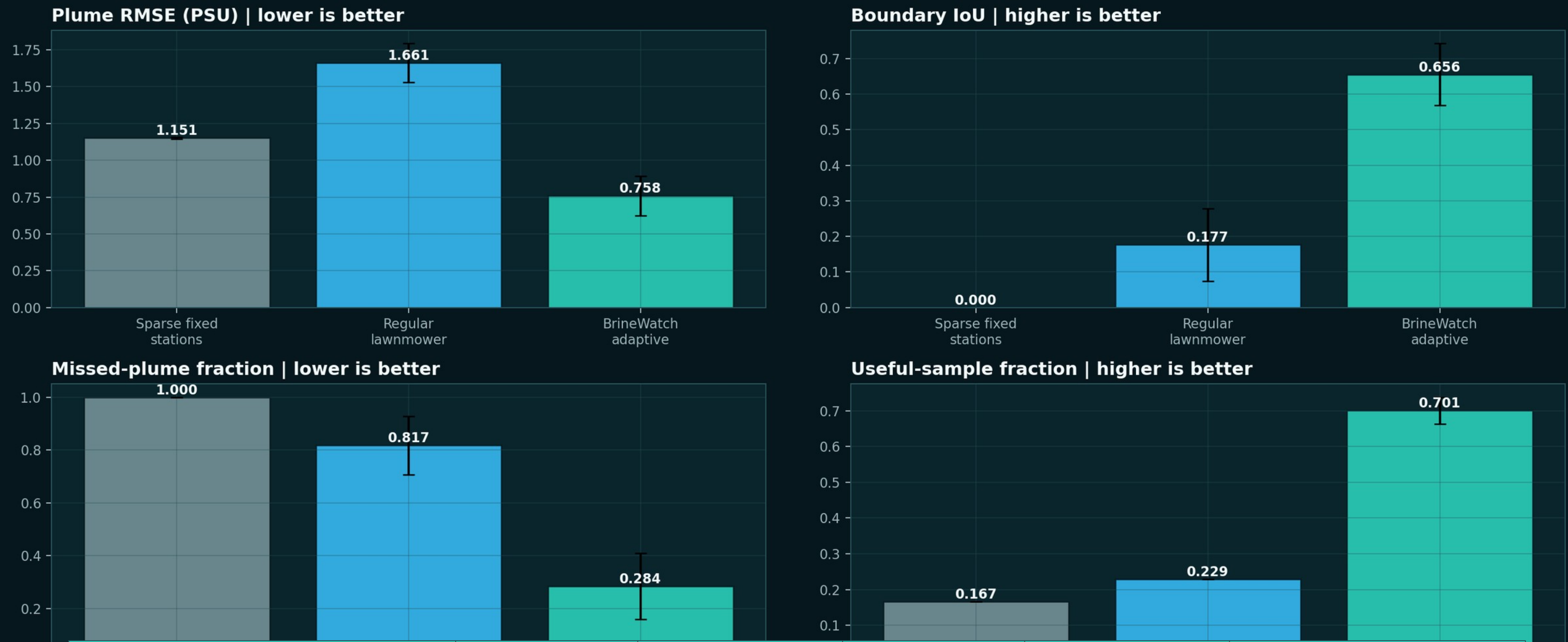
Same 48 readings. More informative spatial evidence.

Eight seeds, same analytic plume, 300 m cap, mission area and CT noise for all three strategies.

EQUAL EVIDENCE BUDGET | 8 SEEDS

Same 48 readings. More informative spatial evidence.

Same analytic plume, 300 m cap, area and noise. Conclusive rate: fixed 0% | lawnmower 50% | adaptive 100%.



Strategy	CLEAR	POSSIBLE	REVIEW	Conclusive	Accuracy if conclusive
Sparse fixed	0 / 8	0 / 8	8 / 8	0%	n/a
Lawnmower	0 / 8	4 / 8	4 / 8	50%	100%
BrineWatch adaptive	0 / 8	8 / 8	0 / 8	100%	100%

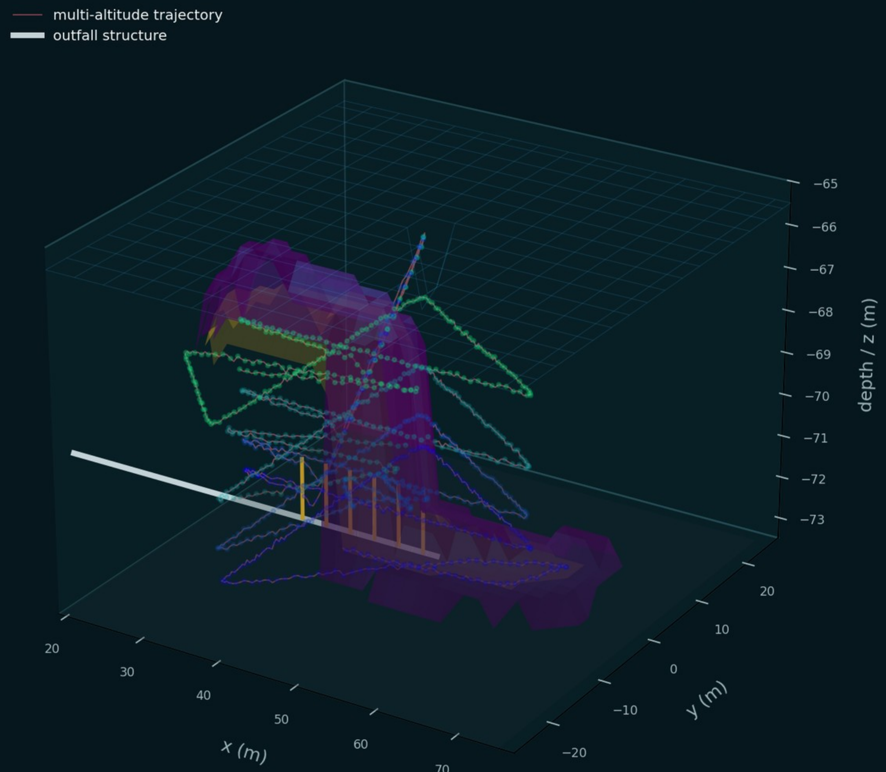
Four altitude bands reveal the plume as a volume

One coherent anisotropic GP connects samples, trajectory, reconstructed isohaline and confidence bounds.

RECONSTRUCT | ONE ANISOTROPIC 3-D GAUSSIAN PROCESS

The plume becomes a volume - with confidence kept visible

Four altitude bands update one coherent GP. Gold is the high-confidence core; cyan wireframe is the possible extent.



0.81
volume IoU

0.477 PSU
3-D RMSE

2051 m3
estimated volume

2448 m3
surrogate truth volume

Simulation surrogate only
not CFD or field truth

912 samples across 0.8, 1.6, 2.8 and 4.5 m bands. 3-D RMSE 0.477 PSU; volume IoU 0.805. Estimated volume 2,051 m3 vs 2,448 m3 surrogate truth - 16.2% under-estimation, replacing the previous over-estimation rather than hiding it.

Conclusive when supported. REVIEW when it is not.

Competition results remain persuasive because limitations and abstentions stay visible.

CLEAR

Correct vs compliant truth

1,049 samples | P(exceed) 0.036 | p95 std 0.83 PSU

REVIEW

Genuinely borderline

290 samples | P(exceed) 0.363 | inconclusive vs compliant truth

POSSIBLE EXCEEDANCE

Correct vs non-compliant truth

591 samples | P(exceed) 1.00 | +1.234 PSU maximum exceedance

What the benchmark reports

Counts and percentages for CLEAR, POSSIBLE EXCEEDANCE and REVIEW; conclusive rate; accuracy among conclusive outputs; false CLEAR; false exceedance; and abstention rate. In the 24-run direct comparison there were zero false CLEAR and zero false exceedance outputs.

Stress evidence kept

The real custom-HoloOcean survey remained REVIEW and reconstructed the difficult field poorly (RMSE 2.72 PSU; boundary F1 0.00). It supports vehicle/sensor integration and safety - not the scientific headline.

NO CERTIFICATION CLAIM

NO CFD CLAIM

NO UNIVERSAL REPLACEMENT CLAIM

The twin is a mission history, not decorative 3-D

A non-expert reviewer can see what changed, how certain it is and what action follows.

BRINEWATCH DIGITAL TWIN | LATEST MISSION

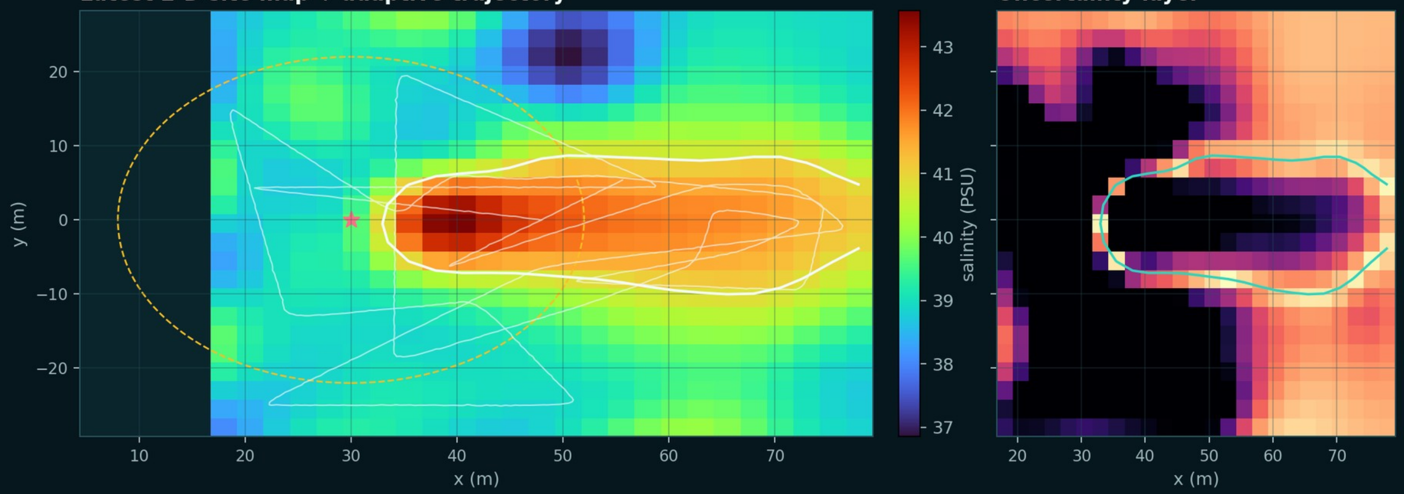
One operational picture, updated mission by mission

Maps, uncertainty, sonar anchor, trajectory, verdict and history remain linked to the evidence ledger.

POSSIBLE EXCEEDANCE	**0.342 PSU**	**0.80**	**0**	**8 / 8**
latest screening	plume RMSE	3-D volume IoU	mission collisions	adaptive correct flags

Latest 2-D site map + adaptive trajectory

Uncertainty layer



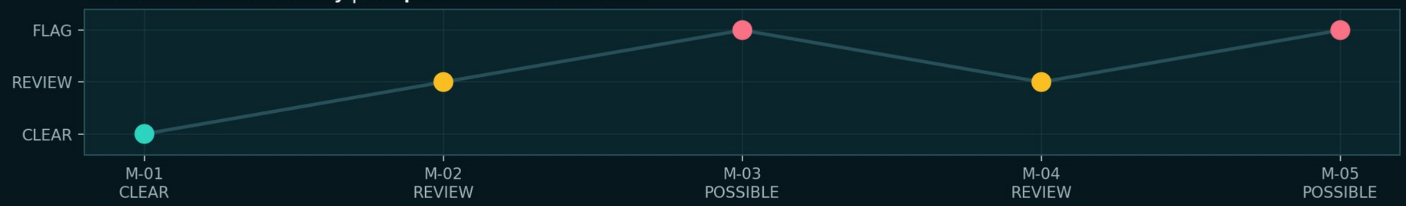
ACT

Escalate targeted reference sampling

Why
P(exceed) = 1.00
Max exceedance = +1.23 PSU
Boundary is spatially resolved

Next waypoint
Highest-risk boundary cell

Simulation mission history | comparable stored evidence



TWIN CONTENTS

sonar anchor | 2-D + 3-D maps
uncertainty | route | verdict
mission history | recommended action

Stored together: latest site map, 3-D plume estimate, uncertainty, ROV trajectory, sonar localization, screening result, previous missions and the recommended follow-up. This is currently simulation-based and not yet connected to a field data service.

Reuse existing hardware to lower the next gate

Ranges are transparent planning assumptions, not vendor quotations or guaranteed savings.

FEASIBILITY | TRANSPARENT PLANNING RANGES

Reuse the robot. Add the sensing. Repeat the evidence.

The team already owns the BlueROV2 and Omniscan 450 FS. Value is strongest for repeated, shore/small-boat-compatible surveys.

FULL NEW BUILD

$22k - 44k$

ROV + 450 FS + CT + integration
calibration + spares + compute

INCREMENTAL NEXT STEP

$9k - 23k$

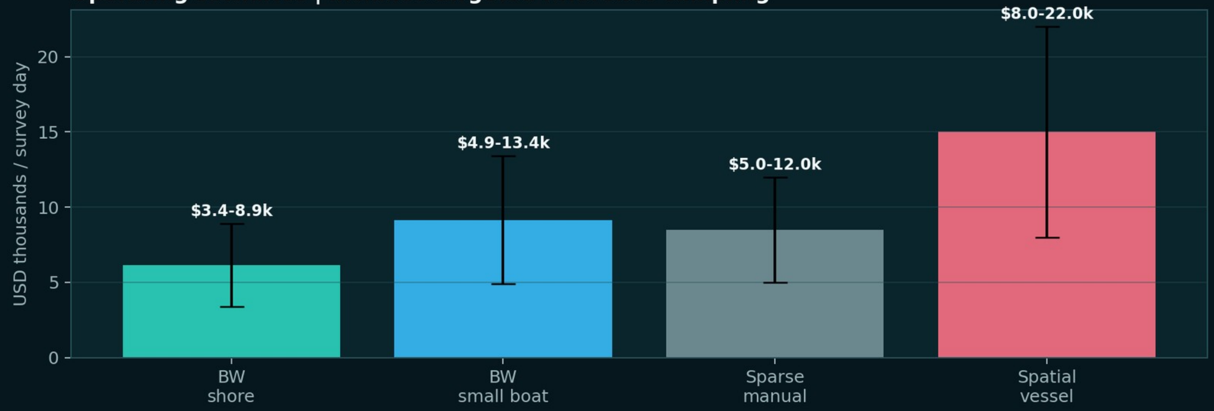
Given owned ROV + sonar
CT payload is the largest open item

INDICATIVE BREAK-EVEN

~2 - 7 repeats

Midpoint scenarios only
not a guaranteed saving

Operating scenarios | includes targeted reference sampling



WHEN IT WORKS

- repeated screening campaigns
- shore or small-workboat access
- owned robotic hardware reused
- same-day map supports targeted follow-up
- inspection and screening share one mobilization

WHEN SAVINGS MAY DISAPPEAR

If vessel, permit and accredited sampling requirements remain unchanged, the value is better spatial evidence - not lower cost.

Assumptions: 2 operators | 2.5-4 h on water | 12 missions/year | 5-year amortization | USD, excluding VAT/duties/contingency. Validate with local quotes.

Public anchors: BlueROV2 base from USD 4,900 (Blue Robotics); Omniscan 450 FS from USD 2,490 (Cerulean Sonar); miniCT-class capability from Teledyne Valeport. CT price, vessel, permits and local labour require quotations. Sources accessed 22 Jul 2026.

Four gates turn a simulation result into field evidence

Autonomy expands only after calibrated truth and supervised operating procedures.

0-3 MONTHS

CT integration

Mount, power, RS232/RS485 interface, time sync, calibration fixture.

Gate: traceable sensor stream

3-6 MONTHS

Controlled water

Known salinity gradients, repeated sonar geometry and mapping runs.

Gate: error bounds vs measured truth

6-12 MONTHS

Nearshore pilot

Permitted, supervised trials with independent reference samples.

Gate: agreement + safe SOP

12-18 MONTHS

Repeated campaign

Site history, operator handover, maintenance and unit economics.

Gate: operational acceptance

Principal risks and mitigations

- **Simulation-to-water gap**
  tank truth + independent reference sampling
- **Surrogate plume physics**
  measured currents, calibrated transport model, later CFD coupling
- **Sonar ambiguity**
  multi-pass confirmation and REVIEW on disagreement
- **Tether and navigation**
  human abort, geofence, standoff and launch plan

A convincing prototype with a precise next gate

Integrate the CT payload. Validate in controlled water. Then target certified sampling.

LOCALIZATION	IN-ENGINE	FLAGSHIP 2-D	VOLUME
2.35 m	**0**	**0.342**	**0.81**
multi-radius centre error	collisions over 395 m	PSU plume RMSE	3-D IoU

What support unlocks

A calibrated CT integration, controlled-water truth campaign and first supervised nearshore pilot. The team already brings the BlueROV2, Omniscan 450 FS, custom HoloOcean scene, mission software, strong demo outputs and a transparent evidence ledger.

One-line value

More informative spatial evidence under constrained survey time - so certified sampling can go where it matters.

STATUS: INITIAL PROTOTYPE OR MODEL

All performance values are simulation results. BrineWatch is not field validated or certified for regulatory compliance.